

Laboratory 3

Fracture Testing

1 Introduction

Fracture is a leading concern in mechanical design and analysis. The purpose of this experiment is to investigate the brittle fracture of polymeric specimens under bending and predict the fracture toughness of the material.

The theory of fracture that is relevant to the particular case of interest is outlined in DOWLING Section 8.4 (specifically, Figure 8.13). Please re-study this section before coming to the lab.

2 Experiment

The experimental setup consists of up to four specimen types with varying geometrical parameters and crack lengths. The specimens will be incrementally loaded until the critical load P_c at which fracture occurs is identified. Record the specimen data below:

Specimen	1	2	3	4
Support distance				
Width				
Thickness				
Crack length				
P_c				

Report your findings by addressing the following questions:

1. Calculate the fracture toughness K_{Ic} for each specimen. Show your calculations.
2. The assistants will also collect the data from each group. We will then post a file on MOODLE containing this data. For each specimen type, carry out a simple statistical analysis by calculating (i) the mean value and (ii) the standard deviation for K_{Ic} .
3. Explain possible reasons for why there is a scatter in the K_{Ic} values for each specimen type.
4. Comment on the quality of your experimental measurements in view of the statistical data.
5. Are the mean K_{Ic} values expected to be the same for each specimen type? Why or why not?
6. In this experiment, is brittle fracture a reasonable assumption? Why or why not?